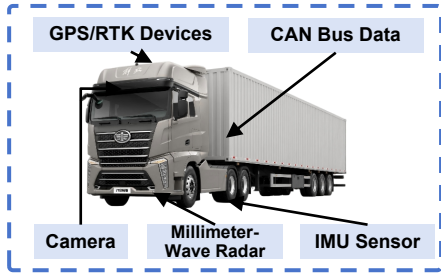
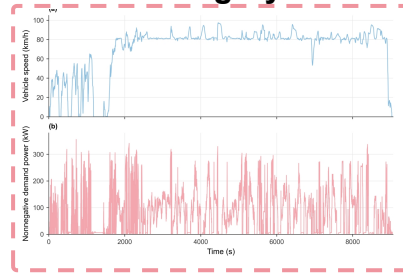


Data Collection



Driving Cycle



Historical Power Window

$$\hat{P}(t-14), \dots, \hat{P}(t)$$

Auxiliary Feature Sequence

Driving Features

Acceleration

Speed

Throttle

Physics-Augmented Features

$P_{(drag)}$

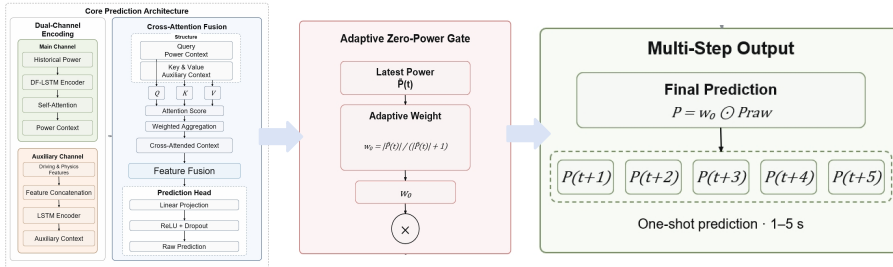
$P_{(roll)}$

$P_{(acc)}$

$P_{(phy)}$

$\Delta P_{(real)}$

Power Prediction

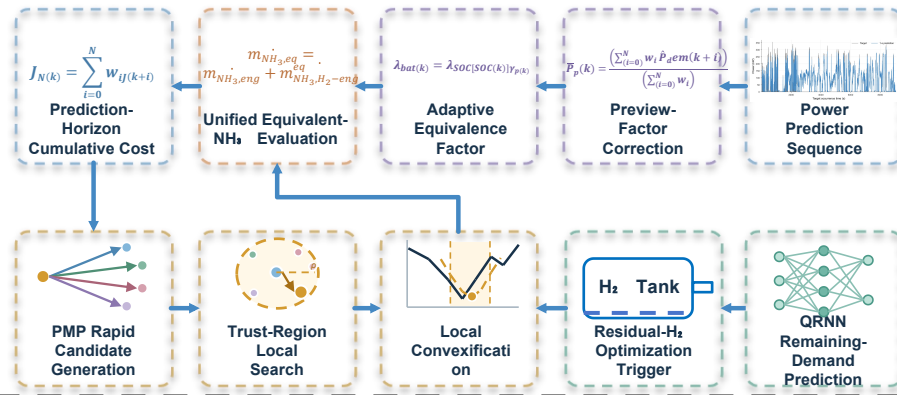


Fuel Conditioning System State Machine

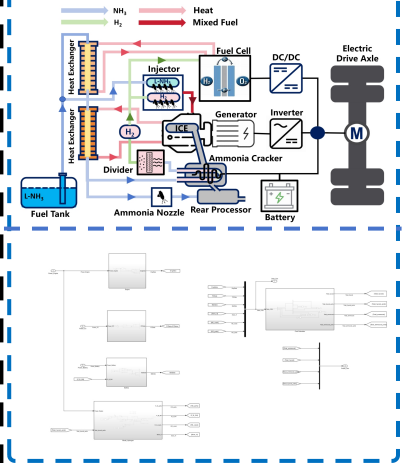
Coupling Variables

Powertrain State Machine

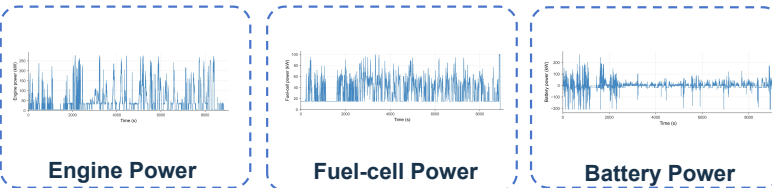
Improved MPC-AECMS with QRNN-Based Residual- H_2 Optimization



Dual-Parallel Finite-State-Machine Constraint System



Distributed Power



System Parameters

